UAPOML Summer Project: Week 1 Assignment

Theoretical Questions

Question 1

The monthly prices are: 100, 108, 103, 115, 110.

(a) Simple Returns: Calculated as $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$

- Month 1: $(108 - 100)/100 = 0.08$ or 8.00%
- Month 2: $(103 - 108)/108 \approx -0.0463$ or -4.63%
- Month 3: $(115 - 103)/103 \approx 0.1165$ or 11.65%
- Month 4: $(110 - 115)/115 \approx -0.0435$ or -4.35%

(b) Log Returns: Calculated as $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

- Month 1: $\ln(108/100) \approx 0.0770$ or 7.70%
- Month 2: $\ln(103/108) \approx -0.0474$ or -4.74%
- Month 3: $\ln(115/103) \approx 0.1103$ or 11.03%
- Month 4: $\ln(110/115) \approx -0.0445$ or -4.45%

(c) Approximation Accuracy: The approximation $r_t \approx R_t$ is least accurate in **Month 3**. This happens because the absolute return is the largest (11.65%). The mathematical reason is that log returns use the Taylor series expansion $\ln(1 + x) \approx x$. This approximation only works well when x is very close to zero; the larger the return, the larger the error.

Question 2

(a) Expected Return of Portfolio:
The linearity-of-expectation property states that the expected value of a sum is simply the sum of the expected values, scaled by their weights: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. Using this rule: $\mathbb{E}[R_p] = 0.4(0.15) + 0.6(0.07) = 0.06 + 0.042 = 0.102$ (or 10.2%).

(b) Impact of Correlation:
No, the result does not change. The linearity of expectation is a universal rule in probability. It holds true whether the assets are independent or highly correlated. Correlation only changes the portfolio's risk (variance), not its expected return.

Question 3

Starting with the basic variance formula:

$$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y) \tag{1}$$

Let $X = w_1R_1$ and $Y = w_2R_2$. Substitute these into the formula:

$$Var(R_p) = Var(w_1R_1) + Var(w_2R_2) + 2Cov(w_1R_1, w_2R_2) \tag{2}$$

We know that constants pull out of variance as squares ($Var(aX) = a^2Var(X)$) and out of covariance normally ($Cov(aX, bY) = abCov(X,Y)$):

$$Var(R_p) = w_1^2Var(R_1) + w_2^2Var(R_2) + 2w_1w_2Cov(R_1, R_2) \tag{3}$$

Finally, we substitute the definitions of variance ($Var(R_i) = \sigma_i^2$) and covariance ($Cov(R_1, R_2) = \rho_{12}\sigma_1\sigma_2$) to get the final formula:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2 \tag{4}$$

Question 4

Given: $w_A = 0.5$, $w_B = 0.5$, $\sigma_A = 0.25$, $\sigma_B = 0.12$.

(a) Portfolio Risk (σ_p):

First, let's plug in the numbers to simplify the variance formula:
$\sigma_p^2 = (0.5)^2(0.25)^2 + (0.5)^2(0.12)^2 + 2(0.5)(0.5)\rho(0.25)(0.12) = 0.019225 + 0.015\rho$

- For $\rho = +1$: $\sigma_p = \sqrt{0.034225} = 0.185$ or 18.5%
- For $\rho = 0$: $\sigma_p = \sqrt{0.019225} \approx 0.1387$ or 13.87%
- For $\rho = -1$: $\sigma_p = \sqrt{0.004225} = 0.065$ or 6.5%

(b) Weighted Average Proof:

The portfolio risk equals the weighted average when $\rho = +1$.
Proof: If $\rho = 1$, the variance formula becomes a perfect square:
$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\sigma_A\sigma_B = (w_A\sigma_A + w_B\sigma_B)^2$.
Taking the square root gives $\sigma_p = w_A\sigma_A + w_B\sigma_B$.

(c) Zero Risk Weight:

For $\rho = -1$, the formula is $\sigma_p = |w_A\sigma_A - (1 - w_A)\sigma_B|$.
To set risk to 0: $0.25w_A - 0.12(1 - w_A) = 0 \implies 0.37w_A = 0.12 \implies w_A = 12/37 \approx$ 32.43%.

Question 5

When two assets are perfectly correlated ($\rho = 1$), they move in exact lockstep, so the portfolio risk is just the weighted average of the individual risks. However, when $\rho < 1$, the assets don't move together perfectly. This means when one asset drops in value, the other might rise, hold steady, or drop less. In the variance formula, a correlation of less than 1 directly reduces the covariance term. This "canceling out" effect smooths out the overall bumps in the portfolio's returns, meaning your total risk ends up being less than the simple sum of its parts.

Question 6

(a) Covariance Matrix Calculation ($\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$):

- $\Sigma_{11} = 0.25 \times 0.25 = 0.0625$
- $\Sigma_{22} = 0.12 \times 0.12 = 0.0144$
- $\Sigma_{33} = 0.04 \times 0.04 = 0.0016$
- $\Sigma_{12} = 0.4 \times 0.25 \times 0.12 = 0.012$
- $\Sigma_{13} = 0.1 \times 0.25 \times 0.04 = 0.001$
- $\Sigma_{23} = 0.2 \times 0.12 \times 0.04 = 0.00096$

$$\Sigma = \begin{bmatrix} 0.0625 & 0.012 & 0.001 \\ 0.012 & 0.0144 & 0.00096 \\ 0.001 & 0.00096 & 0.0016 \end{bmatrix} \tag{5}$$

(b) The matrix is symmetric because correlation works both ways (the correlation between Asset 1 and 2 is the same as between Asset 2 and 1). Therefore, the numbers mirror each other across the main diagonal.

Question 7

(a) Sharpe Ratios ($SR = \frac{\mathbb{E}[R_p]-R_f}{\sigma_p}$):

- Portfolio X: $(14-4)/22 \approx 0.454$
- Portfolio Y: $(9-4)/10 = 0.500$
- Portfolio Z: $(20-4)/35 \approx 0.457$

(b) Portfolio Ranking: Y ¿ Z ¿ X.
A rational investor would prefer Portfolio Y. The Sharpe Ratio measures how much extra return you get for the risk you are taking. Portfolio Y gives the best "bang for your buck" in terms of risk-adjusted return.

(c) Why Y over Z?
Even though Z has a massive 20% return, it takes on a lot of risk to get there. Because Portfolio Y is mathematically more efficient, an investor could simply borrow money at the risk-free rate and invest it into Portfolio Y (leveraging). This allows them to scale up their returns to 20% while keeping their risk lower than Portfolio Z's 35%.

Question 8

(a) Global Minimum Variance (GMV):
The GMV portfolio is the specific mix of assets that gives the absolute lowest possible risk. On a chart where the x-axis is risk, the GMV is the leftmost point because it's impossible to create a portfolio with a lower standard deviation.

(b) Student's Argument:
The student is **incorrect**. The lower half of the curve is called "inefficient" for a reason. For any portfolio on the bottom half, you can look straight up to the top half of the

curve and find a portfolio with the exact same risk but a better return. You should always choose from the top half, no matter how risk-averse you are.

(c) Capital Market Line (CML):

The CML is drawn by starting a line at the risk-free rate on the y-axis and drawing it straight so that it just barely touches (is tangent to) the efficient frontier curve. The slope of this line represents the highest possible Sharpe Ratio available in the market.

Question 9

(a) Standard Error:

$SE = \frac{\sigma}{\sqrt{N}} = \frac{0.30}{\sqrt{36}} = \frac{0.30}{6} = 0.05$ or 5%.

A 5% standard error is huge for monthly returns. This means that if you only rely on historical data, your estimates are mostly just statistical noise and luck, making them very unreliable for the future.

(b) Error Maximisation & Machine Learning:

Mean-Variance Optimisation is an "error maximiser" because it blindly trusts the inputs. If an asset had a lucky historical run, the optimizer assumes it's a great asset and allocates heavily into it. Machine Learning helps fix this by finding actual underlying patterns and relationships, stripping out the historical noise, and giving the optimizer cleaner, more realistic return predictions to work with.

(c) Covariance Parameters:

For 50 assets, the number of unique parameters is $50 \times 51/2 = 1,275$.

This is a major challenge because you need a massive amount of historical data to estimate 1,275 moving parts accurately. If you don't have enough data, the math breaks down and the matrix becomes unstable.